

Group Theory Notes

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Preface

Well as of now haven't yet understood the power of group theory in physics. Here I have tried to understand group theory without dumping the Mathematical Abstractness.

Chapter 1

Representation Theory

1.1 What is Representation?

- A homomorphic map from a group G to group of operators $D(G)$ is called *representation of group G in representation space L*
- dimensionality of representation (n) is dimensionality of vector space.
- let $R, S \in G$ then

$$D(RS) = D(R)D(S) \quad (1.1)$$

$$D(R^{-1}) = [D(R)]^{-1} \quad (1.2)$$

$$D(E) = 1 \quad (1.3)$$

Which follows from the group property and homomorphism of map.

- If $D(G)$ are linear then $\rightarrow$ *linear representation*

-

Example 1.1. Consider the group Z_3 with Cayley table

	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

We'll find the regular representation in following way:

We take the ordered basis $\{|0\rangle, |1\rangle, |2\rangle\}$ and

$$D(g) |h\rangle = |g + h \pmod{3}\rangle.$$

Now the matrix elements can be found out by $D_{ij}(g) = \langle i | D(g) | j \rangle$

So for ex: $D_{11}(0) = \langle 1 | D(0) | 1 \rangle = \langle 1 | 1 \rangle = 1$

$D_{12}(1) = \langle 1 | D(1) | 2 \rangle = \langle 1 | 0 \rangle = 0$

$$D(0) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad D(1) = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad D(2) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}.$$

Check that these matrices follow the group property

- If the mapping is isomorphism, then the order of $D(G)$ is equal to order of group G

1.1.1 Equivalent Representation and Characters

- If we change basis by some matrix C ,

$$D'(R) = CD(R)C^{-1} \quad (1.4)$$

then $D(R)$ and $D'(R)$ are called *equivalent representations*

- Quantity which remain invariant under this is trace

$$\begin{aligned} [D'(R)]_{ii} &= C_{ik} D(R)_{kl} C_{li}^{-1} \\ &= C_{li}^{-1} C_{ik} D(R)_{kl} \\ &= \delta_{lk} D(R)_{kl} \\ &= D(R)_{kk} \end{aligned}$$

(summation over repeated indices to be followed)

- $\text{trace}[D(R)]$ is called *character* of representation D

$$\chi(R) = D_{ii}(R) \quad (1.5)$$

How to construct Representations?

- Given transformation $T : x' = Tx$, associate T with linear operator O_T which acts on functions $\psi(x)$ so that

$$O_T \psi = \psi' \quad (1.6)$$

$$\begin{array}{ccc} x & \xrightarrow{T} & x' \\ \downarrow & & \downarrow \\ \psi(x) & \xrightarrow{O_T} & \psi'(x) \end{array}$$

- This should be such that

$$\psi'(x') = O_T \psi(x') = \psi(x) \quad (1.7)$$

Example 1.2. Suppose T is translation i.e. $Tx = x + a$ then $\psi'(x) = \psi(x - a)$

- $O_T \psi(Ty) = \psi(y)$. Let $Ty = x \implies y = T^{-1}x$ and therefore

$$O_T \psi(x) = \psi(T^{-1}x) \quad (1.8)$$

-

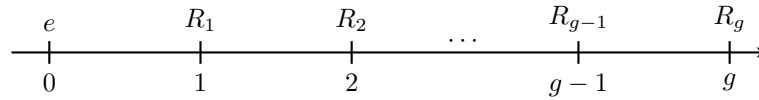
Chapter 2

Lie Groups

2.1 Groups of Infinite Order

- **Idea:** consider the $R_1, R_2, \dots, R_g$ group elements as set of points $\rightarrow$ *Group Manifold*

Example 2.1. label elements by using integers



-
- **Group Table:** defines a function $G \times G \rightarrow G$

$$R_a R_b = R_c \xrightarrow{\text{function}} \phi(a, b) = c$$

Infinite Discrete Groups

2.2 Continuous Groups

- group elements can be parametrized by continuous parameters in a given range
- $\theta_a \rightarrow$ parameters $g(\theta_a) \rightarrow$ Group Element
- We demand: small change of parameter $\rightarrow$ small change in product¹

¹It gives feeling of closeness, continuity

Example 2.2.

$$x' = ax + b$$

is example of finite parameter (2 parameters) continuous group.²

- requirements of continuous group

1. $R(0)R(a) = R(a)R(0) = R(a)$
2. $R(a)R(a^{-1}) = R(a^{-1})R(a) = R(0)$
3. $R(c) = R(a)R(b) \quad c_k = \phi(a_1, a_2, \dots, a_r, b_1, b_2, \dots, b_r)$

- requirements of a *Lie Group*

1. Continuous Group
2. $g(\theta_a)$ is analytic³
3. $R_a^{-1} = h(\theta_a)$ is analytic

- we choose

$$g(0) = 1 \tag{2.1}$$

- We also define generator T_a as

$$\left. \frac{\partial g}{\partial \theta_a} \right|_{\theta_a=0} = -iT_a \tag{2.2}$$

- for small θ_a we can write using Taylor series expansion

$$g(\theta_a) = 1 - i\theta_a T_a \tag{2.3}$$

- now to get an arbitrary group element divide each θ_a in n small parts, each part θ_a/n small, then we can write

$$g(\theta_a) = \left(1 - \frac{i\theta_a}{n} \right)^n \tag{2.4}$$

Taking limit $n \rightarrow \infty$ we get

$$g(\theta_a) = e^{-i\theta_a T_a} \tag{2.5}$$

- if

²If domain of parameters is finite $\rightarrow$ *closed manifold*

³possess derivatives of all orders in all parameters

2.3 Algebra

- Group Multiplication Table defines group
- Not possible to create one for continuous groups
- Group Multiplication can be found out by using Baker-Campbell-Hausdorff formula:

$$e^A e^B = \exp \left(A + B + \frac{1}{2} [A, B] + \frac{1}{12} ([A, [A, B]] + [B, [B, A]]) + \dots \right) \quad (2.6)$$

So we have,

$$e^{-i\theta_a T_a} e^{-i\theta'_b T_b} = \exp \left(-i\theta_a T_a - i\theta'_b T_b + \frac{1}{2} [\theta_a T_a, \theta'_b T_b] + \dots \right) \quad (2.7)$$

- and the commutators satisfy

$$[T_a, T_b] = i f_{abc} T_c \quad (2.8)$$

- The set of all generators constitute *Algebra*⁴. With:
 1. Addition in vector space of generators
 2. Multiplication as the commutator
- If the algebra is define over the Lie Group it is called *Lie Algebra*
- Algebra does not contain all information about the group.
Two different groups can have same algebra

2.4 Differential Representation

- Differential operator, one for each generator, on an infinite dimensional vector space.⁵
- To find differential operators:
 1. Find functions invariant to group transformations (say T). If $x' = Tx$, then invariant function satisfy

$$\phi'(x') = \phi(x) \quad (2.9)$$

We can also write it as

$$\phi'(x) = \phi(T^{-1}x) \quad (2.10)$$

⁴Algebra is ring whose elements form vector space

⁵Operator on a vector space is mapping between two vector spaces

⁶For Eg: Suppose ϕ denotes temperature in a room then changing coordinate system should not change temperature at a point but the functional form of ϕ

2. Consider infinitesimal transformation to get information of algebra

$$x' = x + \delta x \quad (2.11)$$

$$\phi'(x + \delta x) = \phi(x) \quad (2.12)$$

$$\text{or} \quad (2.13)$$

$$\phi'(x) = \phi(x - \delta x) \quad \text{using (2.10)} \quad (2.14)$$

$$\phi'(x) = \phi(x) - \delta x_i \frac{\partial \phi}{\partial x_i} \quad (2.15)$$

3. The group elements will act on these functions now, for an element close to identity we'll have

$$\phi'(x) = (1 - i\delta\theta_a T_a)$$

with

$$T_a = -i \frac{\partial(\delta x_i)}{\partial(\delta\theta_a)} \frac{\partial}{\partial x_i} \quad (2.16)$$

Chapter 3

Rotations and Translations

3.1 Translation Group

- We define translation in multidimensional space as

$$x_i \rightarrow x'_i = x_i + a_i \quad (3.1)$$

- denote this translation by group element t_a . Now obviously

$$t_a t_b = t_{a+b} \quad (3.2)$$

- let us call generators of translation P_i ¹. Hence, we can write

$$t_a = e^{-ia_i P_i} \quad (3.3)$$

- Now, since the translations will commute, the algebra will be abelian

$$[P_i, P_j] = 0 \quad (3.4)$$

3.1.1 Differential Representation

-

$$\begin{aligned} f'(x) &= f(x - a) \\ (1 - iaP + \dots)f(x) &= f(x) - a \frac{d}{dx} f(x) + \dots \end{aligned}$$

This tells us that

$$Pf(x) = -i \frac{d}{dx} f(x) \quad (3.5)$$

- If there are translations in more than one direction, then

$$P_i = -i \frac{\partial}{\partial x_i} \quad (3.6)$$

¹This is momentum in Physics which is generator of translations

3.1.2 Unitary Matrix Representation

3.1.3 Non-Unitary Matrix Representation

- 2×2 representation which satisfies

$$R_a R_b = R_{a+b} \quad (3.7)$$

is

$$\begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix} \quad (3.8)$$

- Similarly 3×3 matrix representation is

$$\begin{pmatrix} 1 & a & a^2 \\ 0 & 1 & 2a \\ 0 & 0 & 1 \end{pmatrix} \quad (3.9)$$

3.2 Rotation Algebra

- Transformations that leave

$$x_1^2 + x_2^2 + x_3^2 = \text{constant} \quad (3.10)$$

- also write it as

$$x_i x_i = \text{constant} \quad (3.11)$$

- x_i 's transforms as

$$x'_i = R_{ij} x_j \quad (3.12)$$

and

$$x'_i x'_i = x_i x_i \quad (3.13)$$

- For infinitesimal transformation we can write

$$R_{ij} = \delta_{ij} + \omega_{ij} \quad (3.14)$$

put this is (3.13)

$$\begin{aligned} (\delta_{ik} + \omega_{ik})(\delta_{il} + \omega_{il}) x_k x_l &= x_i x_i \\ (\delta_{ik}\delta_{il} + \delta_{ik}\omega_{il} + \omega_{ik}\delta_{il}) x_k x_l &= x_i x_i \\ \omega_{kl} x_k x_l + \omega_{lk} x_k x_l &= 0 \end{aligned}$$

which tells us that

$$\omega_{kl} + \omega_{lk} = 0 \quad (3.15)$$

- hence ω 's are antisymmetric and in 3d will have $\binom{3}{2} = 3$ of them and a generator $\mathcal{J}_{ij}$ corresponding to them.

- we can now write a general rotation ²

$$R = \exp \left(-\frac{1}{2} i \omega_{ij} J_{ij} \right) \quad (3.16)$$

- since ω_{ij} is antisymmetric

$$J_{ij} = -J_{ji} \quad (3.17)$$

3.2.1 Differential Representation

- we demand in functional form

$$\phi'(x') = \phi(x) \quad (3.18)$$

- now consider infinitesimal transformations

$$\begin{aligned} \phi'(x) &= \phi(x - \delta x) \\ &= \phi(x) - \delta x_i \frac{\partial \phi}{\partial x_i} \end{aligned}$$

- we have $\delta x_i = \omega_{ij} x_j$ and therefore,

$$\begin{aligned} \phi'(x') &= \phi(x) - \omega_{ij} x_j \frac{\partial \phi}{\partial x_i} \\ &= \phi(x) - \frac{1}{2} \omega_{ij} \left(x_j \frac{\partial}{\partial x_i} - x_i \frac{\partial}{\partial x_j} \right) \phi \\ &= \phi(x) - \frac{1}{2} i \omega_{ij} L_{ij} \phi(x) \end{aligned}$$

where

$$L_{ij} = i \left(x_i \frac{\partial}{\partial x_j} - x_j \frac{\partial}{\partial x_i} \right) \quad (3.19)$$

are generators of rotations

3.2.2 Algebra

- to study algebra, we first find the product

$$\begin{aligned} L_{ij} L_{kl} \phi(x) &= - (x_i \partial_j - x_j \partial_i) (x_k \partial_l - x_l \partial_k) \phi \\ &= - (x_i \partial_j - x_j \partial_i) (x_k \partial_l \phi - x_l \partial_k \phi) \\ &= - [x_i (\delta_{jk} \partial_l \phi + x_k \partial_j \partial_l \phi) - x_i (\delta_{jl} \partial_k \phi + x_l \partial_j \partial_k \phi) \\ &\quad - x_j (\delta_{ik} \partial_l \phi + x_k \partial_i \partial_l \phi) + x_j (\delta_{il} \partial_k \phi + x_l \partial_i \partial_k \phi)] \end{aligned}$$

²Note that the factor of $\frac{1}{2}$ is not merely by convention since there is a summation over i and j , all the terms would be counted twice, like $\omega_{12} \mathcal{J}_{12}$ and $\omega_{21} \mathcal{J}_{21}$

- then the commutator will be

$$[L_{ij}, L_{kl}] = -i(\delta_{ik}L_{jl} - \delta_{il}L_{jk} - \delta_{jk}L_{il} + \delta_{jl}L_{ik}) \quad (3.20)$$

- The fundamental generators J_{ij} should also satisfy the same commutation relation

$$[J_{ij}, J_{kl}] = -i(\delta_{ik}J_{jl} - \delta_{il}J_{jk} - \delta_{jk}J_{il} + \delta_{jl}J_{ik}) \quad (3.21)$$

- in practice in 3-d, we use different notation by defining

$$J_{ij} = -\epsilon_{ijk}\mathcal{J}_k \quad (3.22)$$

and therefore

$$J_{12} = -\mathcal{J}_3 \quad J_{23} = -\mathcal{J}_1 \quad J_{31} = -\mathcal{J}_2 \quad (3.23)$$

- to obtain $\mathcal{J}_k$'s we invert the relation (3.22)

$$\begin{aligned} \epsilon_{ijl}J_{ij} &= -\epsilon_{ijl}\epsilon_{ijk}\mathcal{J}_k \\ (\delta_{jj}\delta_{lk} - \delta_{jk}\delta_{lj})\mathcal{J}_k &= -\epsilon_{ijl}J_{ij} \\ \mathcal{J}_k &= -\frac{1}{2}\epsilon_{ijl}J_{ij} \end{aligned} \quad (3.24)$$

- Therefore, the commutation relations now take form

$$\begin{aligned} [\mathcal{J}_i, \mathcal{J}_j] &= \frac{1}{4}\epsilon_{ipq}\epsilon_{jrs}[J_{pq}, J_{rs}] \\ &= -\frac{1}{4}i\epsilon_{ipq}\epsilon_{jrs}(\delta_{pr}J_{qs} - \delta_{ps}J_{qr} - \delta_{qr}J_{ps} + \delta_{qs}J_{pr}) \\ &= -\frac{1}{4}i(\epsilon_{piq}\epsilon_{pjs}J_{qs} - \epsilon_{piq}\epsilon_{prj}J_{qr} - \epsilon_{qpi}\epsilon_{qjs}J_{ps} \\ &\quad \epsilon_{qpi}\epsilon_{qrj}J_{pr}) \\ &= -\frac{1}{4}i(-J_{ji} - J_{ji} - J_{ji} - J_{ji}) \\ &= -iJ_{ij} \\ &= i\epsilon_{ijk}\mathcal{J}_k \end{aligned} \quad (3.25)$$

3.3 Properties of Rotation Group

- start looking for matrix representation of the group

$$R_{ik}R_{il}x_kx_l = x_ix_i \quad (3.26)$$

$$\implies R_{ik}R_{il} = \delta_{kl} \quad (3.27)$$

- in terms of matrices;

$$\mathbb{R}^T \mathbb{R} = \mathbb{1} \quad (3.28)$$

hence, the matrices are orthogonal;

- The group is thus orthogonal group in three dimensions and is called $S(3)$ ³
- The *Special* comes from the following property

$$(\det \mathbb{R})^2 = 1 \quad (3.29)$$

$$\implies \det \mathbb{R} = \pm 1 \quad (3.30)$$

³SO: Special Orthogonal

Chapter 4

Unitary Groups and Their Representations

4.1 No of Parameter in $U(N)$

- $N \times N$ matrices with complex entries will have $2N^2$ parameters
- we have constrain

$$U^\dagger U = \mathbb{1} \quad (4.1)$$

$$(U_{ij})^* U_{ik} = \delta_{jk} \quad (4.2)$$

- for $j = k$ this tell that the columns of the matrix should be orthonormal, there are N columns giving N constrains
- for $j \neq k$ we can choose j and k in $N(N-1)/2$ ways and this sum for each case should be equal to zero, both real and imaginary parts hence giving us $2 \times N(N-1)/2$ constrains
- total number of constrain then becomes

$$2N^2 - N - N(N-1) = N^2 \quad (4.3)$$

- from (4.1) we get

$$\begin{aligned} \det(U^\dagger U) &= 1 \\ [\det(U)]^* [\det(U)] &= 1 \\ |\det(U)|^2 &= 1 \end{aligned} \quad (4.4)$$

- we can now write

$$\det(U) = e^{i\alpha} \quad (4.5)$$

- we also now write for some arbitrary matrix V

$$U = DV \quad (4.6)$$

with

$$U = e^{i\alpha/N} \mathbb{1} \quad (4.7)$$

and

$$\det(V) = 1 \quad (4.8)$$

- Matrices V also form a group called $SU(N)$ ¹
- the other part is of form of numbers $e^{i\theta}$ which is like ordinary multiplication, we call this group as $U(1)$
- We can write group $U(N)$ as

$$U(N) = SU(N) \times U(1) \quad (4.9)$$

- since $U(1)$ takes one parameter to represent $SU(N)$ has $N^2 - 1$ parameters

4.2 Representation of $U(1)$

- $U(1)$ has one parameter; hence only one generator
- $U(1)$ is an abelian group
- defined by number of form $e^{-i\theta}$, where

$$0 \leq \theta < 2\pi \quad (4.10)$$

- we can write the representation in general form by

$$R(\theta) = e^{-im\theta} \quad (4.11)$$

with $m \in \mathbb{Z}$

- case where $m = 0$ is trivial representation
- we can also define the group with parameter θ taking values in range $0 \leq \theta < A$ by

$$R(\theta) = e^{im\theta} \quad (4.12)$$

with the condition that

$$e^{imA} = 1 \quad (4.13)$$

and therefore

$$m = \frac{2\pi n}{A} \quad n \in \mathbb{Z} \quad (4.14)$$

¹SU:Special Unitary

4.3 Representation of $SU(2)$

- Unitary matrices with determinant 1

$$U = \begin{pmatrix} a & b \\ -b^* & a^* \end{pmatrix} \quad (4.15)$$

with a and b satisfying

$$|a|^2 + |b|^2 = 1 \quad (4.16)$$

- To find the generators of the group we write using constrain equation (4.16); $a_1^2 + a_2^2 + b_1^2 + b_2^2 = 1$ and therefore,

$$a = \sqrt{1 - a_2^2 - b_1^2 - b_2^2} + ia_2 \quad (4.17)$$

$$b = b_1 + ib_2 \quad (4.18)$$

- if all these parameters a_2, b_1 and b_2 are zero, then we obtain the identity matrix. Therefore, we keep these parameters small to obtain an infinitesimal transformation and hence the generators.

$$\begin{aligned} U &= \begin{pmatrix} 1 + ia_2 & b_1 + ib_2 \\ -b_1 + ib_2 & 1 - ia_2 \end{pmatrix} \\ &= \mathbb{1} + ia_2 \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} + ib_1 \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + ib_2 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ &= \mathbb{1} + ia_2 \sigma_z + ib_1 \sigma_y + ib_2 \sigma_x \end{aligned} \quad (4.19)$$

- hence Pauli matrices can be used as generators in fundamental representations

$$[\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma_k \quad (4.20)$$

Chapter 5

Representations of Lorentz Group

5.1 Definition

- Consider the Lorentz Transformation

$$\begin{aligned}x' &= \gamma(x - vt) \\y' &= y \\z' &= z \\t' &= \gamma(t - \frac{v}{c^2}x)\end{aligned}\tag{5.1}$$

where $\gamma = \frac{1}{\sqrt{1-v^2/c^2}}$

- Minkowski realized these transformations leave

$$x_0^2 - x_1^2 - x_2^2 - x_3^2 = \text{constant}\tag{5.2}$$

where $x_0 = ct$

- The Lorentz group contains all transformations that obey (5.2)
- The group is called $O(3, 1)$ and Algebra is called $SO(3, 1)$
- we write (5.2) as

$$g_{\mu\nu}x^\mu x^\nu = \text{constant}\tag{5.3}$$

where g is 4×4 diagonal matrix

$$g = \text{diag}(1, -1, -1, -1)\tag{5.4}$$

- we write Lorentz transformation using operator Λ as

$$x'^\mu = \Lambda^\mu{}_\nu x^\nu\tag{5.5}$$

- then this implies that

$$\begin{aligned} x^\mu x_\mu &= x'^\mu x'_\mu \\ g_{\mu\nu} x^\mu x^\nu &= g_{\mu\nu} x'^\mu x'^\nu \\ &= g_{\mu\nu} \Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma x^\rho x^\sigma \end{aligned}$$

This tells us that

$$g_{\rho\sigma} = g_{\mu\nu} \Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma \quad (5.6)$$

- In matrix notation, this translates to

$$g = \Lambda^T g \Lambda \quad (5.7)$$

5.2 Fundamental Representations And Algebra

The generators

- Consider infinitesimal Lorentz Transformations

$$\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu \quad (5.8)$$

- put this in (5.6)

$$\begin{aligned} g_{\mu\nu} (\delta^\mu{}_\rho + \omega^\mu{}_\rho) (\delta^\nu{}_\sigma + \omega^\nu{}_\sigma) &= g_{\rho\sigma} \\ g_{\rho\nu} \omega^\nu{}_\sigma + g_{\mu\sigma} \omega^\mu{}_\rho &= 0 \\ \omega_{\rho\sigma} + \omega_{\sigma\rho} &= 0 \end{aligned} \quad (5.9)$$

- We now define the generators of the group $J_{\rho\sigma}$ so that the elements of the group can be written as

$$\exp \left(-\frac{1}{2} i \omega^{\rho\sigma} J_{\rho\sigma} \right) \quad (5.10)$$

- since there are $\omega_{\rho\sigma}$ is an anti-symmetric, there will be six of them and hence six generators for the group

Fundamental Representation

- Representation whose states are given by x^μ

$$x'^\mu = \left(\mathbb{1} - \frac{1}{2} i \omega^{\rho\sigma} J_{\rho\sigma} \right)^{\mu\nu} x_\nu \quad (5.11)$$

- keep this to first order in ω and put in

$$x'^{\mu} = x^{\mu} + \omega^{\mu\nu} x_{\nu} \quad (5.12)$$

$$\left(\mathbb{1} - \frac{1}{2} i \omega^{\rho\sigma} J_{\rho\sigma} \right)^{\mu\nu} x_{\nu} = x^{\mu} + \omega^{\mu\nu} x_{\nu}$$

which gives us

$$-\frac{1}{2} i \omega^{\rho\sigma} (J_{\rho\sigma})^{\mu\nu} = \omega^{\mu\nu} \quad (5.13)$$

- This equation has to be satisfied for arbitrary values of ω and must be antisymmetric in indices of ω 's; we construct such J using

$$(J_{\rho\sigma})^{\mu\nu} = i (\delta_{\rho}^{\mu} \delta_{\sigma}^{\nu} - \delta_{\rho}^{\nu} \delta_{\sigma}^{\mu}) \quad (5.14)$$

We can lower all indices and write

$$(J_{\rho\sigma})_{\mu\nu} = i (g_{\rho\mu} g_{\sigma\nu} - g_{\rho\nu} g_{\sigma\mu}) \quad (5.15)$$

- We now get the generators

$$\mathbb{J}_{01} = \begin{pmatrix} 0 & -i & 0 & 0 \\ -i & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad \mathbb{J}_{23} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -i \\ 0 & 0 & i & 0 \end{pmatrix}$$

Chapter 6

Rings

6.1 Introduction

6.1.1 Why Ring?

When we study real number or other structures such as matrices, we generally have two operations "addition" and "multiplication". The ring provides a similar construction.

6.1.2 Definition

A *Ring* is a set R with two binary operations, Addition ($a+b$) and Multiplication (ab) such that $\forall a, b, c \in R$:

1. $a + b = b + a$
 2. $(a + b) + c = a + (b + c)$
 3. Additive Identity 0; $a + 0 = a$
 4. Additive Inverse $-a$; $a + (-a) = 0$
 5. $a(bc) = (ab)c$
 6. $a(b + c) = ab + ac$ and $(b + c)a = ba + ca$
- If Commutative under multiplication $\implies$ *Commutative Ring*
 - a is called *unit* of ring if a^{-1} exist
 - **Examples of Rings:**
 1. $\mathbb{Z}$ under ordinary addition and multiplication. Units of ring are 1 and -1. No other than this as for eg. $2 \in \mathbb{Z}$ but $2^{-1} \notin \mathbb{Z}$
 2. $\mathbb{Z}_n$ under addition and multiplication modulo n .

Theorem 6.1. *Ring Multiplication Rules* $a, b, c \in R$

1. $a0 = 0a = 0$
2. $a(-b) = (-a)b = -(ab)$
3. $(-a)(-b) = ab$
4. $a(b - c) = ab - ac$ and $(b - c)a = ba - ca$
If R has unit element
5. $(-1)a = -a$
6. $(-1)(-1) = 1$

Proof:

1. $a + a0 = a = a(0 + 0) = a0 + a0$ Now canceling one terms from both side gives $a0 = 0$. Now, $0 + 0a = 0a = (0 + 0)a = 0a + 0a \implies 0a = 0$

2.

$$\begin{aligned}
 a + (-a) &= 0 \\
 ab + (-a)b &= 0 \\
 ab + (-a)b - (ab) &= -(ab) \quad \{\text{add } -(ab) \text{ both sides}\} \\
 (-a)b &= -(ab)
 \end{aligned}$$

3.

$$\begin{aligned}
 (a + (-a))(b + (-b)) &= 0 \\
 ab + -ab - ab + (-a)(-b) &= 0 \\
 -ab + (-a)(-b) &= 0 \\
 (-a)(-b) &= ab
 \end{aligned}$$

4. $a(b + (-c)) = ab + a(-c) = ab - ac$

5.

$$\begin{aligned}
 a(1 + (-1)) &= 0 \\
 a + (-1)a &= 0 \\
 (-1)a &= -a
 \end{aligned}$$

6. Trivial from property 3

Chapter 7

SU(2)

- The SU(2) Algebra is given by

$$[J_i, J_j] = i\epsilon_{ijk}J_k \quad (7.1)$$

7.1 J_3 Eigenstates

- Our goal is to completely reduce the Hilbert Space to block-diagonal form
- But since here nothing commutes with anything else, we can only diagonalize one of them (J_3)